Carlos Hernández Martínez

QA & AI Engineer | Backend Developer

Bridging the gap between AI innovation and software reliability.

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SUMMARY

Computer Engineer graduated from the Polytechnic University of Valencia with specialization in Artificial Intelligence and Machine Learning. Professional experience developing and validating AI agents, building QA automation, and working with backend systems using Python, Spring Boot and Spring Batch. At Mercadona IT, he contributes to ATENEA, an AI agent platform, and IntegraT, a Spring Boot integration testing framework.

Core strengths: Research-to-production mindset - End-to-end ownership - Autonomous problem-solving - Cross-disciplinary collaboration - International team communication - Fast ramp-up & self-directed learning

PROFESSIONAL EXPERIENCE

QA & AI Engineer

Jul. 2025 - Present

Mercadona IT

Contributes to two internal projects at Mercadona IT: ATENEA, an AI agent platform built with Google ADK, LiteLLM and Claude, and IntegraT, a Spring Boot integration testing framework. Validates AI agents and develops agents for QA tasks, including generating tests from user stories and automating test cases. Contributes to ongoing work on a framework and methodology for evaluating AI agents, microservices and agent harnesses with tools and skills. Explores access control for skills and MCP server scoping in subagent definitions through an agent definition protocol that extends MCP. Also works on automated test data management, backend integration tests and quality analysis with SonarQube.

Technologies: Python - Spring Boot & Spring Batch - ATENEA - Google ADK - LiteLLM - Claude - AI Agents for QA - AI Agent Evaluation - MCP - Subagent Scoping - User Story to Test Generation - Test Automation - IntegraT - Integration Tests - Mutant Testing - SonarQube

Backend AI Engineer

Oct. 2024 - Jun. 2025

Urobora SL (AI Startup)

Developed core AI infrastructure for an early-stage startup: autonomous agents for task automation, RAG systems for enterprise knowledge management, and microservices architecture on GCP/Kubernetes. Worked end-to-end from research to production deployment.

Technologies: Python - FastAPI - LLMs & RAG - GCP

EDUCATION

Bachelor's Degree in Computer Engineering

2021-2025

Polytechnic University of Valencia (UPV)

Specialization in Artificial Intelligence and Machine Learning. Training in algorithms, data structures, software architectures, and machine learning systems.

Erasmus in Computer Science

6 months, 2024

Technical University of Munich (TUM)

International exchange program focused on AI and robotics. Participation in applied research projects on implicit neural models.

Akademia Bankinter

2023

Bankinter Innovation Foundation

Training program in innovation and technological entrepreneurship. Development of leadership skills and management of innovative projects.

Diploma in Social Innovation 2019-2020

CEU San Pablo Valencia

Training in social innovation and corporate responsibility in collaboration with Edwards Lifesciences.

SKILLS

Backend & Core Engineering
Python - Java - Spring Boot - Spring Batch

Artificial Intelligence
AI Agents Development - Autonomous Agents Validation - Google ADK - LiteLLM - Claude - Model Context Protocol (MCP) - PyTorch - TensorFlow

QA & Software Quality
Integration FWK (Spring Boot) - IntegraT - AI QA Automation - User Story to Test Generation - Mutant Testing - SonarQube - Playwright - Postman

DevOps & Cloud
Google Cloud - CI/CD (GitHub Actions, Jenkins, CloudBees) - Spinnaker

Professional competencies
- Technical Leadership & Mentoring
- Strategic Product Vision
- Cross-functional Collaboration
- Analytical Thinking & Complex Problem Solving
- Adaptability in Agile & Startup Environments
- International Technical Communication

PROJECTS

Medical Diagnosis System with Deep Learning - Bachelor's Thesis

Bachelor's thesis project developing an automatic classification system for knee osteoarthritis using medical radiographs. Implementation of CNN architectures (ResNet, EfficientNet) with PyTorch and TensorFlow, achieving 80% accuracy in binary classification. Application of fine-tuning techniques, data augmentation, and cross-domain validation. Focus on real clinical utility and scalability of the assisted diagnosis system.

Tags: Python - PyTorch - TensorFlow - Computer Vision - Healthcare AI

Links: https://github.com/Tempus23/Radiography_TFG

Neural Implicit Models for Robotics - TUM Research

Collaborative research at the Technical University of Munich (TUM) with KUKA Robotics. Development of implicit neural models for real-time collision detection in robotic manipulators. PyTorch implementation of swept volume models and optimization for efficient inference. Contribution to research paper at the intersection of Deep Learning and industrial robotics.

Tags: Python - PyTorch - Robotics - Research - Neural Networks

Links: https://github.com/Amrsaeed/FastCollisionDetection

LANGUAGES

Spanish: Native - English: B2/C1 (Cambridge Certificate)